

AGONPROJECT

An online platform for exploring approaches to formal argumentation

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Why AgonProject?

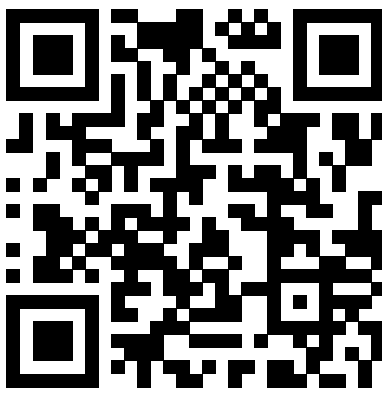
Formal argumentation offers a rich zoo of formalisms for reasoning with conflicting information — but their diverse semantics, algorithms, and implementation details make them hard for students and newcomers to grasp.

TWEETYPROJECT provides the most comprehensive collection of argumentation implementations available, yet exposes only *text-based* interfaces for a handful of tasks.

Goal: a unified, **visual**, install-free web interface to *construct*, *evaluate*, and *compare* argumentation formalisms — right in the browser.

Built with Vue 3 + TypeScript (client-side single-page app); reasoning tasks are delegated to a TWEETYPROJECT REST backend. Open source under GPL-3.0.

Try it now — no installation



agon.tweetyproject.org

One Platform — Six Argumentation Formalisms

AgonProject

The platform to explore different approaches to formal argumentation.

Abstract Argumentation

The foundational model for the formal representation of argumentation via arguments and directed attacks.

OPEN EXAMPLE

meal_wine two_cycles grounded_ideal initial_sets

NEW

Create new Generate random AF

Bipolar Argumentation

Relations between arguments can be either attacking or supporting.

OPEN EXAMPLE

meal_wine vacation_planning murder_trial

NEW

Create new Generate random BAF

Dialectical Argumentation

Relations between arguments are modelled via propositional acceptance conditions.

OPEN EXAMPLE

meal_wine murder_trial

NEW

Create new Generate random ADF

Incomplete Argumentation

Distinguish between certain and uncertain arguments and attacks to represent incomplete information.

OPEN EXAMPLE

murder_trial

NEW

Create new Generate random IAF

Probabilistic Argumentation

Arguments and attacks can be assigned a probability value between 0 and 1 to quantify uncertainty.

OPEN EXAMPLE

murder_trial

NEW

Create new Generate random PAF

Argumentation with Collective Attacks

Extends abstract argumentation by allowing sets of arguments to collectively attack a target argument.

OPEN EXAMPLE

tandem_trip

NEW

Create new

AF *Abstract* argumentation — arguments and directed attacks (Dung).

BAF *Bipolar* — attacks **and** a support relation.

ADF *Abstract dialectical* — propositional acceptance conditions.

iAF *Incomplete* — uncertain arguments and attacks.

PAF *Probabilistic* — probabilities on arguments and attacks.

SetAF *Collective attacks* — sets of arguments attack a target.

Every formalism shares the same editor, evaluation, export, generation, and learning features below — so comparing them is effortless.

What You Can Do

Build

Interactive graph editor with directed, circular, radial and force-directed layouts (Graphviz) plus a physics simulation.

Evaluate

Extension- and ranking-based semantics; credulous / sceptical acceptance; highlight extensions directly in the graph; compare in parallel windows.

Export & Share

One-click $\text{\LaTeX}$ (argumentation package), SVG, TGF, and ICCMA AF-format — plus a shareable URL.

Generate

Random frameworks from eight networkX generators (Erdős–Rényi, Watts–Strogatz, Barabási–Albert, ...).

Learn

10+ step-by-step tutorials, an extensive glossary with literature references, and curated example frameworks.

In Action

Extensions Preferred - Enumerate

Ranking Categoriser

Evaluate & rank (AF). Preferred extensions are enumerated while the *categoriser* ranking is annotated directly on each argument; evaluation windows can stay open in parallel.

Models Stable - Enumerate

Acceptance conditions (ADF). Each argument carries a propositional acceptance condition; AGONPROJECT evaluates three-valued interpretations and highlights a stable model in the graph.

Export

Collective attacks & export (SetAF). A set of arguments jointly attacks a target, drawn as hyperedges; the export window emits ready-to-use $\text{\LaTeX}$ with an SVG preview.

Supported Semantics

Group	Semantics
Classical	conflict-free, admissible, complete, grounded, preferred, stable
Admissibility-based	ideal, semi-stable, eager, strong admissibility, initial sets, unchallenged
Non-admissible	naive, CF2, SCF2, stage, stage2, (strongly) undisputed
Weak-admissible	w-admissible, w-complete, w-grounded, w-preferred
Rankings	burden, categoriser, counting, discussion, iterated graded defense, serialisability, social, strategy, tuples
Available across AF, BAF, iAF, PAF and SetAF; ADFs use the three-valued classical and naive semantics. Serialisation sequences are supported interactively.	

Outlook

Planned extensions:

- further formalisms — Assumption-Based Argumentation, Weighted AFs, Bipolar Set-based AFs, claim-augmented AFs;
- benchmark generators from the ICCMA ecosystem;
- explanation-based reasoning: *why* is an argument accepted?

AGONPROJECT is backend-agnostic and open to new formalisms and solvers.

Live: agon.tweetyproject.org • Source: github.com/aig-hagen/AgonProject • GPL-3.0

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